NAME: Kishan pandey

CUID :24250183

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# Experiment No. 9
# Object Detection and Recognition using Deep Learning Models

from ultralytics import YOLO
import cv2

# Load a pre-trained YOLO model
# The model will be downloaded automatically during the first run
model = YOLO("yolov8n.pt")

# Load an image
image_path = "image.jpg"
image = cv2.imread(image_path)

# Check whether the image was loaded
if image is None:
    print("Error: image.jpg was not found.")
    print("Place image.jpg in the same folder as this Python file.")
    exit()

# Perform object detection
results = model(image)

# Display detected objects
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for result in results:

    # Print detected class names and confidence scores
    if result.boxes is not None:
        for box in result.boxes:
            class_id = int(box.cls[0])
            confidence = float(box.conf[0])
            class_name = model.names[class_id]

            print(
                f"Object: {class_name}, "
                f"Confidence: {confidence:.2f}"
            )

    # Draw bounding boxes and labels
    annotated_image = result.plot()

# Display the result
cv2.imshow("YOLO Object Detection", annotated_image)

# Save the output image
cv2.imwrite("detected_objects.jpg", annotated_image)

print("Output saved as detected_objects.jpg")
print("Press any key to close the image window.")

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Observation

YOLO detects multiple objects in an image using bounding boxes and class labels.

Confidence scores indicate the model's confidence in each prediction.

The pre-trained model can detect common objects from the COCO dataset.

Detection performance may be affected by image quality, object size, occlusion,

lighting conditions, and background complexity.

Question 1: Differentiate between image classification, object detection, and image segmentation.

Image classification assigns one or more labels to an entire image.

Object detection identifies objects using bounding boxes and class labels.

Image segmentation assigns a class label to individual pixels or regions.

Question 2: Explain the working principle of the YOLO algorithm.

YOLO processes the complete image in a single forward pass through a neural network.

It predicts bounding boxes, class probabilities, and confidence scores simultaneously.

This makes YOLO suitable for real-time object detection.

Question 3: Compare YOLO, SSD, and Faster R-CNN based on speed, accuracy, and applications.

YOLO is generally fast and suitable for real-time applications.

SSD is also a single-stage detector with a balance between speed and accuracy.

Faster R-CNN is a two-stage detector and usually provides high accuracy but is slower.

Question 4: What is a bounding box? Why is it important?

A bounding box is a rectangle surrounding a detected object.

It shows the object's location and helps separate it from the background.

Question 5: Explain confidence score and IoU.

A confidence score represents how confident the model is about a detection.

Intersection over Union measures the overlap between predicted and actual bounding boxes.

Higher IoU indicates better localization.

Question 6: What are the advantages of pre-trained deep learning models?

Pre-trained models reduce training time, require less data, and provide strong features.

They are generally more effective than manually designed image-processing rules.

Question 7: Why are MS COCO and Pascal VOC widely used?

They contain many images, object categories, annotations, and bounding boxes.

They provide standard benchmarks for evaluating object detection models.

Question 8: Mention five real-world applications.

Applications include autonomous driving, surveillance, healthcare, retail analytics,

and industrial inspection.

Question 9: What challenges occur in complex environments?

Challenges include occlusion, poor lighting, small objects, overlapping objects,

motion blur, cluttered backgrounds, and changes in object appearance.

Question 10: How can object detection be improved?

Performance can be improved using better datasets, data augmentation, transfer learning,

larger models, hyperparameter tuning, improved feature extraction, and model pruning.

OUTPUT

